STRUCTURAL NOTES ON $L/2L$ FOR *AN ORDER ON THE LEECH LATTICE FROM A $\mathbb{Z}_3$-SYMMETRIC TRIPLE-OCTONION PRODUCT* SUBALGEBRA AND IDEAL STRUCTURE OF THE MOD-2 QUOTIENT

OPENCODE DEEPSEEK, HERMES [1] (PROPOSED DISCOVERIES) AND CLAUDE OPUS 4.7 [2] (VERIFICATION)

Abstract. The main paper's Appendix A, Section A.1 ("The mod-2-quotient reading of Tables A.2 and A.3") stops at the structural statement: under the descended octonion product on $\bar{L} := L/2L$, the subspace $V := \sigma(L\bar{s})/2L$ is a 4-dimensional left ideal and $W := \sigma(Ls)/2L$ is a 4-dimensional subalgebra, with $\bar{L} = V \oplus W$.

External feedback from *OpenCode DeepSeek, Hermes* proposed two finer structural descriptions: an explicit basis and multiplication table for W (with a 3-dim nilpotent ideal and a universal zero divisor), and an explicit basis for V together with total isotropy under the natural quadratic form on $\bar{L}$. We reproduce both proposals verbatim, then record the empirical verification carried out in this project (`python_project/src/verify_discovery1_W_subalgebra.py` and `python_project/src/verify_discovery2_V_isotropic.py`).

Of the two proposals, one (Discovery 1, the W multiplication table) required two cell corrections and one structural re-interpretation; the other (Discovery 2, V totally isotropic) is correct in its asserted form and is sharpened here in two new directions: V is in fact a *two-sided* ideal of $\bar{L}$ (not only a left ideal, as in the paper), and W is also totally isotropic under the natural quadratic form (correcting the proposed "mixed isotropy" claim). Together, V and W form a Witt decomposition of $\bar{L}$ into a pair of maximal totally isotropic subspaces (Lagrangians) under the plus-type quadratic form induced by the norm on L.

The Lagrangian-polarization framing developed here is now recorded at the end of Section A.1 of the main paper.

1. Setting

Throughout, $L = D_8^+$ is the E_8 lattice in the coordinate-symmetric placement, $\sigma = (1\ 2)$ is the transposition of two imaginary basis units of the paper, $L\bar{s}$ and Ls are Wilson's sublattices, and $\bar{L} := L/2L \cong \mathbb{F}_2^8$ is the mod-2 quotient with the descended (Z-bilinear $\to$ $\mathbb{F}_2$-bilinear) octonion product $\bar{\mu}$. The paper's Appendix A, Section A.1 establishes:

(1) $\bar{L}$ has an inherited $\mathbb{F}_2$-bilinear product $\bar{\mu}$ whose structure constants are the entries of Table A.1 reduced modulo 2. The product is non-unital ($e_0 \notin L$) and non-commutative in the paper's L-basis $L_1, \ldots, L_8$.
(2) $V := \sigma(L\bar{s})/2L$ and $W := \sigma(Ls)/2L$ are 4-dimensional $\mathbb{F}_2$-subspaces of $\bar{L}$ with $\bar{L} = V \oplus W$.
(3) Lemma 4.3 $\iff$ V is a left ideal of $(\bar{L}, \bar{\mu})$.
(4) Lemma 4.4 $\iff$ W is a subalgebra of $(\bar{L}, \bar{\mu})$.

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This supplement records two structural proposals about the mod-2 quotient $L/2L$ communicated by an external contributor identified as *OpenCode DeepSeek, Hermes*, alongside an empirical verification carried out in this project. Where the proposals were correct or partially correct, the structure is recorded with attribution; where corrections were required, the original claim is reproduced verbatim and corrected forward. The supplement is companion material to the main paper (`paper/main.tex`, v4); it is not part of the paper and is not in its referee scope.

The proposals below refine (3) and (4). All claims have been verified by exact-arithmetic enumeration on the paper's Appendix-A basis; verification scripts are in `python_project/src/verify_discovery1_W_subal` and `verify_discovery2_V_isotropic.py`.

2. Discovery 1 (OpenCode DeepSeek, Hermes): the multiplication table of W

Original proposal (verbatim).

> **Discovery 1.** $W = \sigma(Ls)/2L$ – explicit subalgebra identified. I computed the multiplication table of $W = \sigma(Ls)/2L$ as an $\mathbb{F}_2$-algebra. This is the structural reason for Lemma 4.4.
>
> Basis in $L/2L$ coordinates:
>
> $$\begin{aligned} w_0 &= (1,1,1,0,0,0,0,0) \text{ – contains identity of } L/2L \\ w_1 &= (0,1,0,1,1,0,1,0) \\ w_2 &= (0,0,0,1,0,1,1,1) \text{ – universal zero divisor} \\ w_3 &= (0,0,0,0,1,0,1,1) \end{aligned}$$
>
> Multiplication table:
>
> | | w_0 | w_1 | w_2 | w_3 |
> |---|---|---|---|---|
> | w_0 | $w_0 + w_1$ | 0 | w_2 | 0 |
> | w_1 | $w_1 + w_2 + w_3$ | 0 | 0 | w_2 |
> | w_2 | 0 | 0 | 0 | 0 |
> | w_3 | w_3 | w_2 | 0 | 0 |
>
> Structural features:
> - $N = \mathrm{span}\{w_1, w_2, w_3\}$ is a 3-dim nilpotent ideal, $N^2 = \mathrm{span}\{w_2\}$, $N^3 = 0$.
> - w_2 is a universal zero divisor.
> - The identity of $L/2L$ is in W but NOT idempotent in W ($w_0^2 = w_0 + w_1$).
> - Only $6/64$ basis products are non-zero.
>
> This replaces the appendix tables with an abstract structural reason: Lemma 4.4 holds because $W \in \mathrm{Subalg}(L/2L)$, provable structurally rather than by enumerating 64 coefficients.

Verification result. The proposed basis of W is correct: the four vectors (w_0, w_1, w_2, w_3) are in W and span exactly the 4-dimensional $\mathbb{F}_2$-subspace $W = \sigma(Ls)/2L$.

The multiplication table requires two cell corrections, and one structural claim is incorrect; the original framing as an "$\mathbb{F}_2$-algebra in the conventional sense" overstates the structure since the descended product on $\bar{L}$ is non-unital (the octonion identity $e_0 \notin L$) and non-commutative in the L-basis.

Actual multiplication table. Verified by direct computation of $\bar{\mu}(w_i, w_j)$ using the structure constants in the paper's Appendix-A L-basis:

$\bar{\mu}(\cdot,\cdot)$	w_0	w_1	w_2	w_3
w_0	w_0	0	w_2	0
w_1	w_1	0	0	w_2
w_2	0	0	0	0
w_3	w_3	w_2	0	0

Six non-zero entries; every non-zero entry is a single basis vector.

Identity / idempotent claim. In L-basis coordinates, $w_0 = L_1 + L_2 + L_3$. Since $L_2 + L_3 = (e_0 + e_1) + (e_0 - e_1) = 2e_0 \in 2L$, this reduces modulo $2L$ to $w_0 \equiv L_1 = s$. The octonion identity e_0 is not in L (it has odd coordinate sum, so it lies outside D_8^+), and $\bar{L}$ has no multiplicative identity. The corrected statement: w_0 represents s modulo $2L$ and is an *idempotent in* W ($w_0^2 = w_0$), not an inherited identity of $\bar{L}$.

Structural features (corrected confirmations). With the actual table:

- $N := \operatorname{span}_{\mathbb{F}_2}\{w_1, w_2, w_3\}$ is a 3-dimensional two-sided ideal of W.
- $N^2 = \mathbb{F}_2 w_2$ and $N^3 = 0$.
- w_2 is a universal zero divisor of W (every product $\bar{\mu}(w_2, w) = 0$ for all $w \in W$, and $\bar{\mu}(w, w_2)$ is 0 or w_2).
- W decomposes as $W = \mathbb{F}_2 w_0 \oplus N$, with w_0 idempotent.

Does this replace the appendix tables? Not by itself. The mod-2 quotient subsection (paper Section A.1) already converts Lemma 4.4 to "W is a subalgebra of $\bar{L}$"; the table above describes the internal structure of W as an algebra, which is a refinement. Verifying W closes under $\bar{\mu}$ still requires checking $4 \times 4 = 16$ basis products in $\mathbb{F}_2$ — a reduction from the appendix tables' 64 integer products in $\mathbb{Z}$, but not an elimination. A genuine structural argument would identify W intrinsically inside $\bar{L}$ (kernel of a natural map, image of a canonical construction, Peirce component, radical-like piece) without referencing σ at all. That identification is not yet in hand.

3. Discovery 2 (OpenCode DeepSeek, Hermes): V totally isotropic left ideal

Original proposal (verbatim).

> **Discovery 2.** $V = \sigma(L\bar{s})/2L$ is a totally isotropic left ideal. V basis:
>
> $$(0,1,1,0,0,0,0,0),\ (0,0,1,0,1,0,1,0),\ (0,0,0,1,0,0,1,1),\ (0,0,0,0,1,1,0,1).$$
>
> Properties verified:
> - V is a left ideal: $\bar{\mu}(L/2L, V) \subseteq V$ (Lemma 4.3 structurally).
> - V is totally isotropic: $q(v) = 0$ for all $v \in V$.
> - $V \oplus W = L/2L$.
> - W has mixed isotropy (8 with $q = 0$, 8 with $q = 1$).

The quadratic form. For an even unimodular lattice L with norm form $N(v) = \sum v_i^2$, the descended $\mathbb{F}_2$-valued quadratic form

$$q\colon L/2L \to \mathbb{F}_2, \qquad q(v + 2L) := N(v)/2 \pmod 2$$

is well-defined because $N(v+2u) = N(v) + 4\langle v, u\rangle + 4N(u) \equiv N(v) \pmod 4$, so the right-hand side is σ-independent of the chosen representative. For $L = D_8^+$ this q is the standard plus-type form on $\mathbb{F}_2^8$: 136 isotropic vectors (including 0) and 120 anisotropic vectors out of 256, Witt index 4.

Verification result. The proposed basis of V is correct: the four vectors above are in $V = \sigma(L\bar{s})/2L$ and span exactly it.

The two correct claims and the four-by-four scoresheet:

Claim	Verified
V is a left ideal of $\bar{L}$	Yes (already paper Lemma 4.3)
V is totally isotropic, $q \equiv 0$ on V	Yes (all 16 elements)
$V \oplus W = \bar{L}$	Yes (already paper Section A.1)
W has mixed isotropy (8/8)	**No** – W is totally isotropic

W is totally isotropic (correction to proposed mixed isotropy). Direct computation gives $q(w) = 0$ for all 16 elements $w \in W$. This is empirically clean: V and W are *both* 4-dimensional totally isotropic subspaces of $\bar{L}$, and the proposed 8/8 split for W is incorrect under the natural quadratic form $q(v) = N(v)/2 \pmod 2$. (The proposal may have used a different form; under the standard one, W is totally isotropic.)
Bonus structural fact: V is also a right ideal. The paper records V as a left ideal (the mod-2 form of Lemma 4.3). Direct empirical test gives also

$$\bar{\mu}(V, \bar{L}) \subseteq V,$$

so V is in fact a *two-sided ideal* of $\bar{L}$. This is a genuine structural fact not appearing in the paper's mod-2 reading, and the right-ideal direction does not follow from the left-ideal direction (the algebra $\bar{\mu}$ is non-commutative in the L-basis).

The Witt decomposition picture. Putting the verified pieces together: under the plus-type quadratic form q on $\bar{L} \cong \mathbb{F}_2^8$,

$$\bar{L} = V \oplus W, \tag{1}$$

where:

- V is a 4-dimensional *maximal totally isotropic two-sided ideal* of $(\bar{L}, \bar{\mu})$;
- W is a 4-dimensional *maximal totally isotropic subalgebra* of $(\bar{L}, \bar{\mu})$;
- $V \cap W = \{0\}$, and V, W together span $\bar{L}$;
- the Witt index of the plus-type form on $\mathbb{F}_2^8$ is 4, so each of V and W is a maximal isotropic subspace (Lagrangian).

This is a clean orthogonal-geometry picture of what σ accomplishes structurally: a transversal decomposition of $\bar{L}$ into a pair of complementary Lagrangians, one carrying the two-sided-ideal role and the other carrying the subalgebra role.

Observation. *V and W are not only orthogonally placed – they are geometrically related: under the bilinear form $b(x, y) := q(x + y) - q(x) - q(y)$ associated to q, the pairing between V and W is non-degenerate (since their direct sum is $\bar{L}$ and both are isotropic), so V and W are dual Lagrangians of the orthogonal space $(\bar{L}, q)$.*

4. Discovery 1 vs Discovery 2: are the ideals the same?

The user (and a reader) might reasonably ask whether the 3-dim nilpotent ideal $N \subset W$ from Discovery 1 is the same object as the 4-dim two-sided ideal $V \subset \bar{L}$ from Discovery 2. *They are not.* Their relationship is:

- V lives in the complementary summand to W in the decomposition $\bar{L} = V \oplus W$;
- N lives *inside* W (it is a sub-ideal of the subalgebra W), and is therefore disjoint from V except at 0 (empirical: $|N \cap V| = 1$).

The two ideals are structurally distinct components of the same overall mod-2 picture: V is the "ideal half" of the $V \oplus W$ decomposition (it carries the left-multiplication action of $\bar{L}$), and N is the radical-like core of the "subalgebra half" W. Both are totally isotropic under q; V is in addition a two-sided ideal of all of $\bar{L}$, whereas N is only known to be a two-sided ideal of W.

5. Relevance to the paper's open question

The paper's Section 7 records as open: *a structural reason for Lemma* **??**, asking specifically for an "identification of $\sigma(Ls)$ with a left ideal, with the image under a norm-preserving octonion automorphism, or with some module-theoretic structure over L, that would lift the mod-2 picture". The structure documented here is the next layer past Section A.1: not the integer-level structural reason the open question targets, but a sharper account of the mod-2 picture itself — specifically, the Witt-decomposition interpretation and the internal multiplication table of W. These do not close the open question (the integer-level lift remains open) but they refine the target: any future intrinsic identification of $\sigma(Ls)$ should descend to recover the multiplication table of W recorded in §2, with w_0 idempotent and w_2 a universal zero divisor.

Attribution and provenance

Both structural proposals (**Discovery 1** and **Discovery 2**) were communicated by an external contributor identified as *OpenCode DeepSeek, Hermes.* The proposals reached this project through review feedback on the main paper's Appendix A mod-2-quotient subsection; the original wording is reproduced verbatim in §2 and §3.

The empirical verification (existence of V and W with the proposed bases, the multiplication table of W, total isotropy of V, total isotropy of W under the natural quadratic form, the two-sided-ideal property of V, and the structural separation of $N \subset W$ from $V \subset \bar{L}$) was carried out in this project by Claude Opus 4.7 (Anthropic) at the direction of Jens Köplinger, using the scripts `python_project/src/verify_discovery1_W_subalgebra.py` and `verify_discovery2_V_isotropic.py`.

Corrections to the original proposals (the two-cell error in the W multiplication table, the misidentification of w_0 with "the identity of $L/2L$", and the mistaken "W has mixed isotropy" claim) are recorded in §2–§3 with the verbatim originals preserved.

Cross-references

- Paper Appendix A, Section A.1 ("The mod-2-quotient reading of Tables A.2 and A.3"): the level at which the paper's structural account stops.
- Paper §7, open question (1): the integer-level structural reason for Lemma 4.4 (*remains open*).
- `python_project/src/verify_mod2_quotient.py`: paper-side verification of V as a left ideal and W as a subalgebra.
- `python_project/src/verify_discovery1_W_subalgebra.py`: verification of the W-multiplication table (with two-cell corrections to the proposal).
- `python_project/src/verify_discovery2_V_isotropic.py`: verification of V totally isotropic, two-sided ideal, W also totally isotropic.
- `paper/reviews/2026-05-24_claude_opus_4_7_W_concrete.md`: the project-internal record of the W concrete example.
- `prompt_logs/155_W_subalgebra_concrete.txt` and `156_appendix_renumber_discovery_record.txt`: prompt logs.

[1]External contribution

[2]Anthropic, acting at the direction of Jens Köplinger